QQ-C-557B
September 20, 1974
SUPERSEDING
Fed. Spec. QQ-C-557A
June 21, 1963

FEDERAL SPECIFICATION

COPPER-NICKEL ALLOY SHOT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers copper-nickel alloy in the form of shot.

1.2 Classification. Material covered by this specification shall be of the following alloys, as specified (see 6.2):

Alloy A - 70-30 copper-nickel.

Alloy B - 50-50 copper-nickel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-1806 - Barrel and Kegs, Wood, Slack.
PPP-D-705 - Drum: Metal Shipping, Steel (Over 25 and Under 55 Gallon).
PPP-D-723 - Drums, Fiber.

FSC 9630

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Office in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and Standards required by contractors in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Chemical composition. The material shall conform to the chemical requirements specified in table I.

Table I. Chemical composition, percent

Alloy	Copper	Nickel	Copper + nickel, min.	Iron, max.	Manganese, max.	Silicon, max.	Carbon, max.	Sulphur, max.	Lead, max.	Tin, max.	Zinc, max.
A	Rem.	28.5-31.5	99.0	0.7	0.25	0.05	0.05	0.05	0.05	0.05	0.05
B	Rem.	48.5-51.5	98.5	1.0	0.30	0.05	0.05	0.05	0.05	0.05	0.05

3.2 Size. Unless otherwise specified (see 6.2), shot shall range from 1/8 to 3/4 inch in diameter.

3.3 Workmanship. The shot shall be of uniform quality and free from burned metal or foreign inclusions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all shot made from the same melt of copper-nickel alloy. If the melt of a shipment cannot be identified, a lot shall consist of 1000 pounds of shot of the same composition.

4.3 Sampling.

4.3.1 For chemical analysis. At least one representative sample shall be taken from each lot of shot. The sample shall weigh not less than 1 pound.

4.3.2 For visual and dimensional examination. Each lot shall be examined.

4.4 Examination.

4.4.1 Visual and dimensional. Examination shall be made to determine compliance with the requirements for size (see 3.2) and workmanship (see 3.3).

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4.4.2 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Test procedures.

4.5.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1. In case of dispute, analysis by method 111 shall be the basis for acceptance.

4.6 Rejection.

4.6.1 Examination defects. Any lot having one or more defects shall be rejected.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5.1, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2 and 6.3).

5.1.1 Level A. Shot of like size and description shall be packed in fiber drums conforming to PPP-D-723, type II; or in metal drums conforming to PPP-D-705; or in wood barrels conforming to PPP-B-1806, type I or II. The gross weight of each container shall not exceed the weight limitations of the applicable container specification.

5.1.2 Level B. Shot of like size and description shall be packed as specified in 5.1.1 except that fiber drums may be type I.

5.1.3 Level C. The products shall be separated by alloy and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier in accordance with the Uniform Freight Classification or National Motor Freight Classification for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking (see 6.2 and 6.3).

5.2.1 Civil agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Shot covered by this specification is intended for use as a master alloy addition agent for copper casting alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Alloy required (see 1.2).
(c) When size other than that specified herein is required (see 3.2).
(d) Whether material is to be packed by level A, B, or C (see 5.1).
(e) Special marking, if required (see 5.2).
(f) Maximum gross weight of containers.

6.3 The requirements for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Alloy B of this specification is equivalent to ASTM B 358.

MILITARY CUSTODIANS:

Army - MR
Navy - OS
Air Force - None

Review activities:

Army - None
Navy - SH
Air Force - None

User activities:

Army - None
Navy - MC
Air Force - None

Preparing activity:

Army - MR

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price 30 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

QQ-C-557B, Copper-Nickel Alloy Shot

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426

REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE.

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